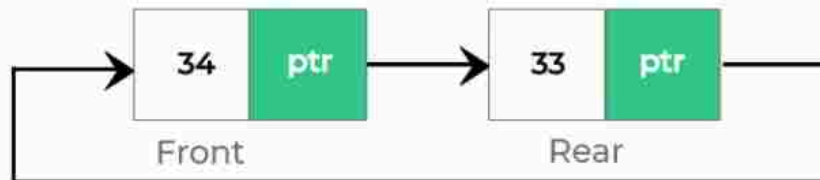


DS	Session	Queue	Question Information	Level 1	Challenge 59
Question description On the last day of the semester, Shahid's students were talking and playing very loudly to the delight of the end of the semester. The principal of the college severely reprimanded Shahid. But he decided to engage them in a different activity without getting angry at the students. So Shahid gave his students to solve the task such that, Circular Queue using Linked List and dequeue two elements from circular queue. there is no memory waste while using Circular Queue, it is preferable than using a regular queue. Because linked lists allow for dynamic memory allocation, they are simple to build. Circular Queue implementation using linked list is identical to circular linked list except that circular Queue has two pointers front and back whereas circular linked list only has one pointer head. Final Year students were lacking the idea to solve the problem. Being an exciting youngster can you solve it? Function Description					
CH.SC.U4CSE25225 - M ASHWIN Department of Computer Science & Engineering (CSE) 2026					

enqueue(34)



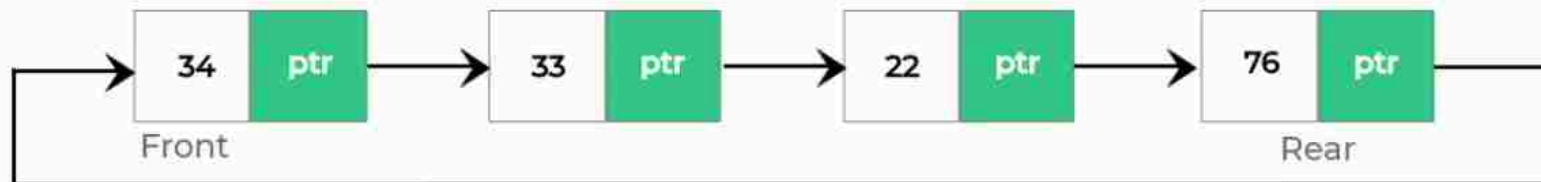
enqueue(33)



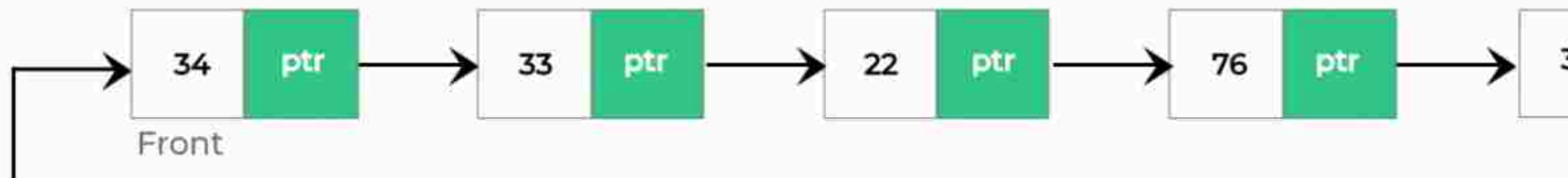
enqueue(22)



enqueue(76)

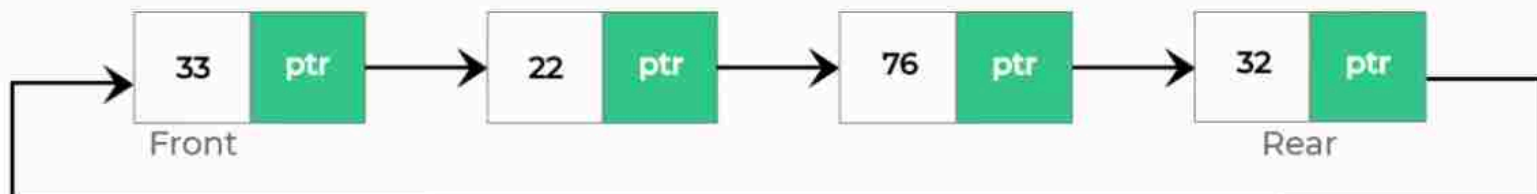


enqueue(32)



Removing the elements from Queue

dequeue()



Constraints:

0 < size < 100

0 < data < 1000

Input format:

First line indicates the size of the queue

Second line indicates the elements of the queue.

Output Format:

First line indicates the inserted elements in queue using linked list concept

second line indicates the one element dequeued from the queue using linked list concept.

Logical Test Cases

Test Case 1

INPUT (STDIN)

5
9 1 5 3 5

EXPECTED OUTPUT

9 1 5 3 5
1 5 3 5
5 3 5

Test Case 2

INPUT (STDIN)

8
1 8 6 4 9 5 3 5

EXPECTED OUTPUT

1 8 6 4 9 5 3 5
8 6 4 9 5 3 5
6 4 9 5 3 5

Mandatory Test Cases

Test Case 1

KEYWORD

void enqueue(int d)

Test Case 2

KEYWORD

struct node* next;

Test Case 3

KEYWORD

n = (struct node*)malloc(sizeof(struct n

Test Case 4

KEYWORD

struct node *f = NULL;

Test Case 5

KEYWORD

struct node *r = NULL;

Test Case 6

KEYWORD

n->data = d;

Test Case 7

KEYWORD

n->next = NULL;

Test Case 8

KEYWORD

struct node* t;

Complexity Test Cases

Test Case 1

CYCIOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

355

Test Case 3

NLOC

75

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node* next;
7 };
8
9 struct node *f = NULL;
10 struct node *r = NULL;
11
12 void enqueue(int d) {
13     struct node* n = (struct node*)malloc(sizeof(struct node));
14     n->data = d;
15     n->next = NULL;
16     if (f == NULL && r == NULL) {
17         f = r = n;
18         r->next = f;
19     } else {
20         r->next = n;
21         r = n;
22         r->next = f;
23     }
24 }
25
```

```
26void dequeue() {  
27    struct node* t = f;  
28    if (f == NULL && r == NULL) {  
29        return;  
30    } else if (f == r) {  
31        f = r = NULL;  
32        free(t);  
33    } else {  
34        f = f->next;  
35        r->next = f;
```

main | CH.SC.U4CSE25225 | Queue9.c

Ln 69, Col 1 Spaces: 4 UTF-8 CRLF C

```
36    free(t);  
37    }  
38}  
39  
40void display() {
```

```
41     struct node* t = f;
42     if (f == NULL && r == NULL) {
43         return;
44     }
45     do {
46         printf("%d ", t->data);
47         t = t->next;
48     } while (t != f);
49     printf("\n");
50 }
51
52 int main() {
53     int size;
54     if (scanf("%d", &size) != 1) return 0;
55
```

```
56     for (int i = 0; i < size; i++) {  
57         int d;  
58         scanf("%d", &d);  
59         enqueue(d);  
60     }  
61  
62     display();  
63     dequeue();  
64     display();  
65     dequeue();  
66     display();  
67     return 0;  
68 }  
69
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5

9 1 5 3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue9.exe

9 1 5 3 5

1 5 3 5

5 3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

8

1 8 6 4 9 5 3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue9.exe

1 8 6 4 9 5 3 5

8 6 4 9 5 3 5

6 4 9 5 3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>